Numerical Solutions of Laplace Equation

Comparison of Steady-State and Transient Methods

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Computational Physics II

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Outline

1. Problem Definition
2. Numerical Methods
3. Numerical Formulation
4. Results

Problem Definition

Solve $\nabla^2 \psi(x,y) = 0$ on $[0,3] \times [0,3]$, subject to:

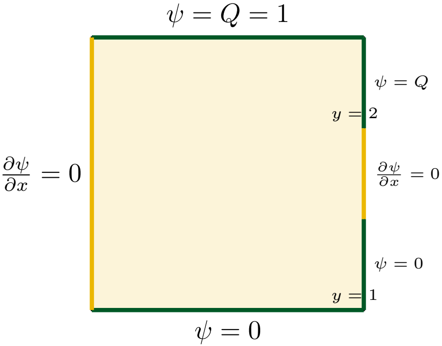
$$\psi(x,3) = Q = 1, \quad 0 \le x \le 3,$$

$$\psi(3,y) = \begin{cases} 1, & 2 \le y \le 3, \\ 0, & 0 \le y \le 1, \end{cases}$$

$$\psi(x,0) = 0, \quad 0 \le x \le 3,$$

$$\frac{\partial \psi}{\partial x}(0,y) = 0, \quad 0 \le y \le 3,$$

$$\frac{\partial \psi}{\partial x}(3,y) = 0, \quad 1 \le y \le 2.$$



Numerical Methods

Steady–State Methods

- **SOR**
  - $\omega^\star = \dfrac{2}{1 + \sin\big(\pi/(N-1)\big)}$
- **Conjugate Gradient**
  - solves the sparse SPD system
  - BCs embedded in the RHS

Transient Methods

- **Forward Euler**
  - CFL: $\Delta t \leq \dfrac{h^2}{4}$
  - first–order in time
- **Runge–Kutta 4**
  - fourth–order in time
  - larger stable step

Numerical Formulation

Finite–Difference Approximation

- Five-point stencil

$$-4\psi_{i,j} + \psi_{i+1,j}+\psi_{i-1,j} + \psi_{i,j+1}+\psi_{i,j-1} = 0$$

- Neumann (2nd-order ghost)

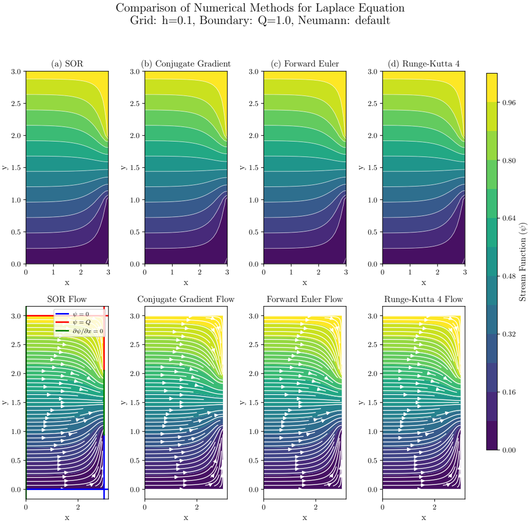
$$\psi_{0,j} \;=\; \tfrac{4}{3}\,\psi_{1,j} - \tfrac{1}{3}\,\psi_{2,j}$$

Matrix Formulation (CG)

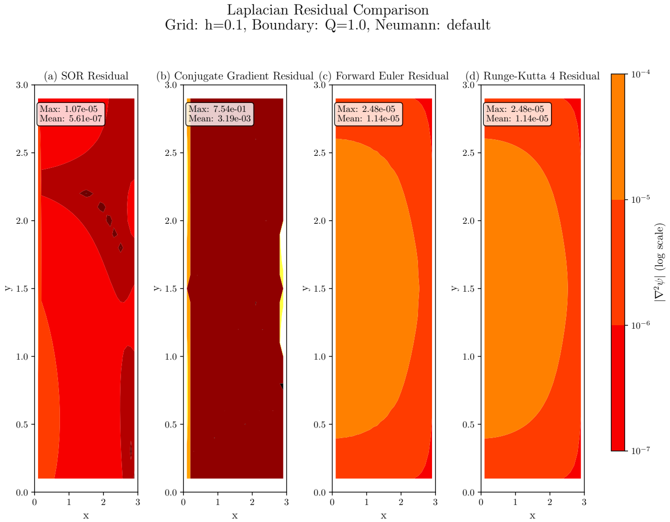
$$A\Psi = b, \qquad A\text{: sparse, SPD}$$

- $-4/h^2$ on the diagonal
- $+1/h^2$ for each neighbour
- Extra $+1/h^2$ on diag at Neumann faces
- Dirichlet / ghost terms $\to b$

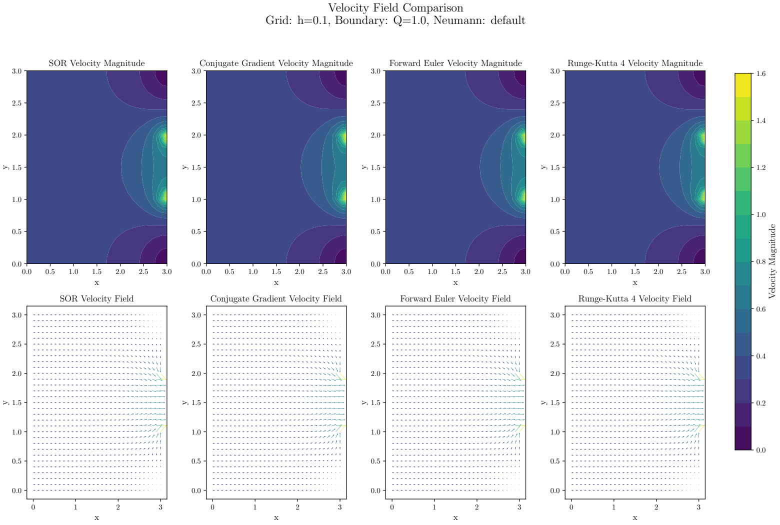
Numerical Methods – default case



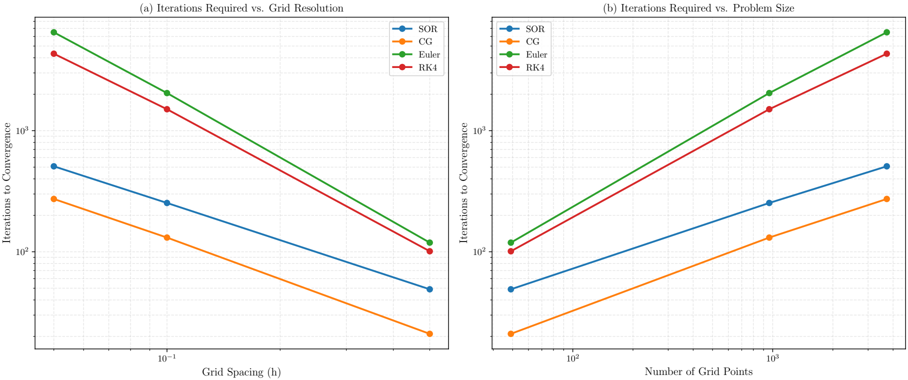
Error Map ($h = 0.1$)



Velocity Field ($h = 0.1$)



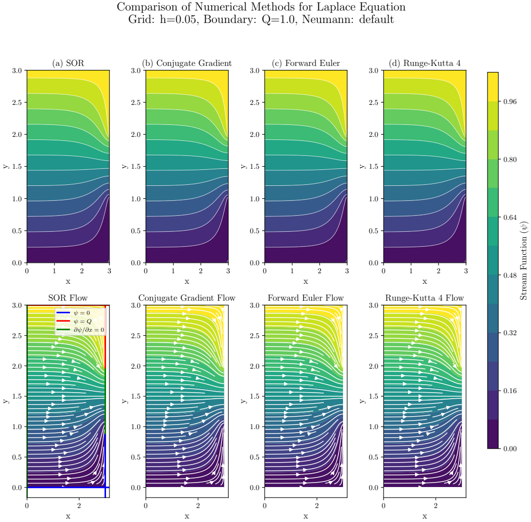
Iterations vs Problem Size



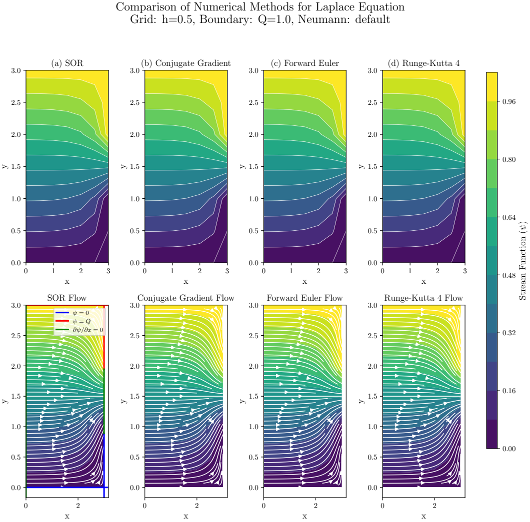
Conclusions

- All four solvers reach the **same steady field** for every boundary mix (Dirichlet, Neumann, or mixed).
- **Who is fastest?**
  CG $<$ SOR ($\omega \approx \omega_{\mathrm{opt}}$) $\ll$ RK4 $<$ Euler.
- All retain the intended $O(h^2)$ **spatial accuracy**; second–order ghost points keep Neumann walls consistent.
- **Take-away**: use CG for production speed, **but** relatively harder to code in.

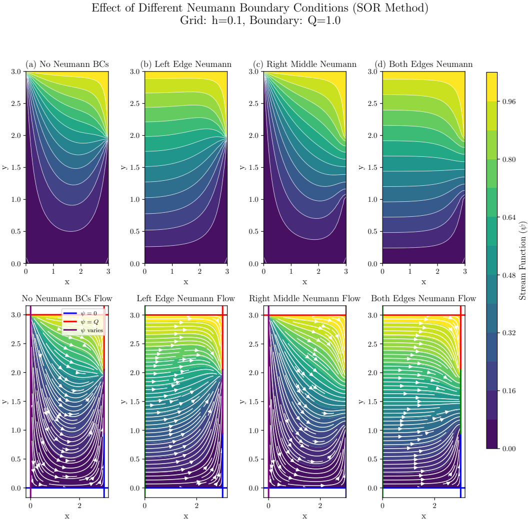
Method Comparison — Fine Grid ($h = 0.05$)



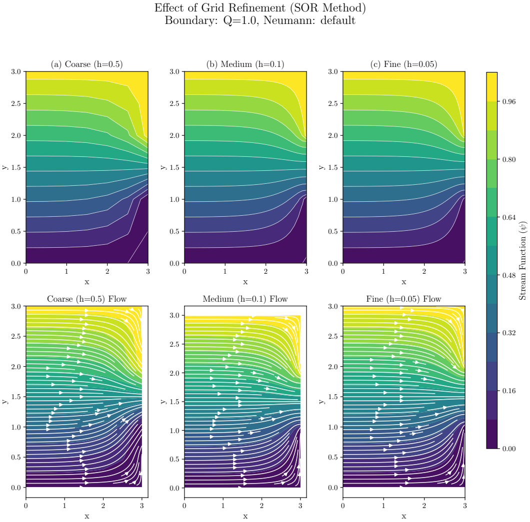
Method Comparison — Coarse Grid ($h = 0.5$)



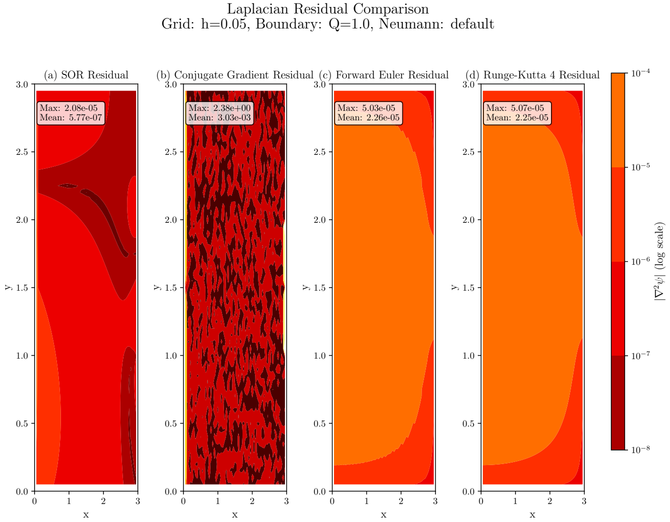
Effect of Neumann-Boundary Placement (SOR)



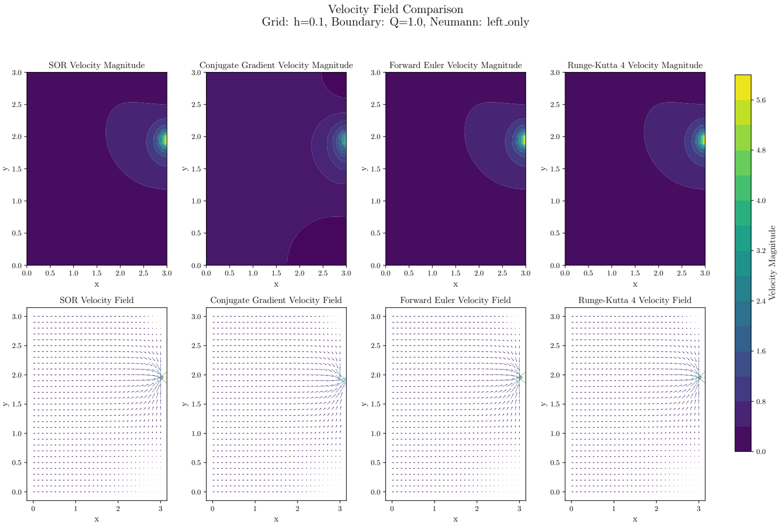
Grid-Refinement Study (SOR)



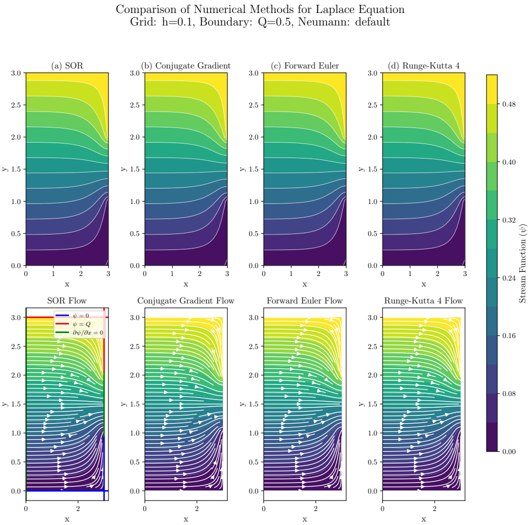
Laplacian Residual Map — Fine Grid



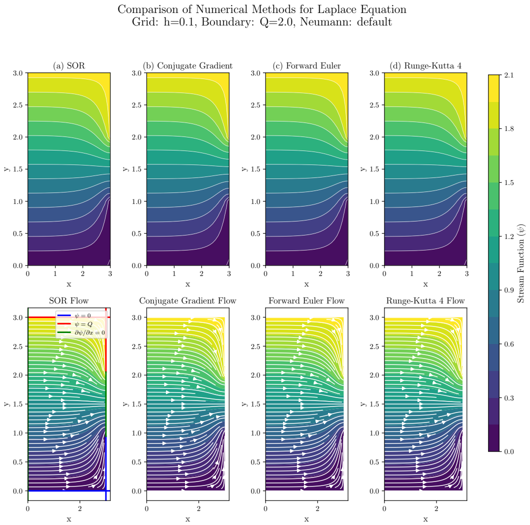
Velocity Field — Left-Edge Neumann Only



Method Comparison — Boundary Value $Q = 0.5$



Method Comparison — Boundary Value $Q = 2.0$



Thank You!

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